

Monotone Classification with Relative Approximations

Lecture Notes for SC2320 Data Analytics and Mining

April 5, 2026

Instructions to readers

The flow of this set of notes is in tandem to the flow of the original paper. It will also go in the same flow as the presentation slides. Concepts/Material/Information that is auxiliary to the core ideas of this paper will be outlined at the end of each section.

Acronym Reference

Acronym	Full Form
ER	Entity Resoluton
EM	Entity Matching
ACR3	Acronym Three

1 Background and Motivation

1.1 The Challenge of Entity Matching

Definition 1.1 (Entity Matching). Given two sets of entities E_1 and E_2 , the task is to determine, for every pair $(e_1, e_2) \in E_1 \times E_2$, whether e_1 and e_2 refer to the same real-world entity. It is a binary classification task that determines whether a candidate pair is a match (+1) or non-match (-1) [2].

To motivate the problem, consider the following scenario where we have two sets of **entities** that we refer to here as product listings - E_1 and E_2 , sourced from two different e-commerce platforms — say Shopee and Lazada respectively. Each listing carries attributes such as **p-name**, **p-description**, **p-year**, and **p-price**. The goal of Entity Matching in this scenario is to decide, for every pair $(e_1, e_2) \in E_1 \times E_2$, if they are referring to the same real-world product.

What makes this non-trivial is that even a genuinely matching pair may disagree on attribute values. For instance, one platform may list a product as “*Sony WH-1000XM5*” while another lists it as “*Sony Noise Cancelling Headphones XM5 Series*”. They are referring to the same product but with different representations. It is therefore insufficient to rely solely on attribute-level comparison to determine a match. To attain maximum accuracy in entity matching, it falls upon a human expert to manually inspect each pair and make a verdict. However, with thousands of listings on each platform, the number of pairs in $E_1 \times E_2$ grows quadratically - making manual inspection labor intensive and expensive. This motivates the need for a more systematic and efficient approach to Entity Matching.

1.2 The Entity Resolution (ER) Pipeline

To address the Entity Matching Problem, we transform it into a classification problem. This is done by passing datasets through an ER Pipeline consisting of three key steps: **Blocking**, **Comparison**, and **Matching**. The first two steps serve as preprocessing stages that progressively reduce and structure the problem, while the third step is what Entity Matching entails and is the primary focus of the paper. We will go over the first two steps in brief before diving into greater detail on the third.

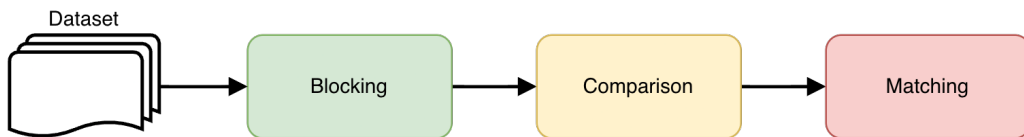


Figure 1: The ER Pipeline.

1.2.1 Blocking

As mentioned earlier in 1.1, the two entity sets P_1 and P_2 can grow quadratically, making it computationally infeasible to compare every pair. The goal of blocking is to narrow the search space down to a smaller candidate subset $S \subseteq P_1 \times P_2$, retaining only pairs that are plausible matches based on an application-dependent heuristic.

Going back to the Shopee-Lazada example, one may restrict S to only those pairs (p_1, p_2) where **p-year** agrees across both listings. Note that blocking is optional — if skipped, $S = P_1 \times P_2$.

Remark 1.1 (Application-Dependent Heuristic). The criterion used for blocking is **application-dependent**; there is no universal rule for what constitutes an “obvious non-match”.

For example:

- **E-commerce:** A mismatched **p-year** is a reliable signal to discard a pair.
- **Hospital records:** Mismatched blood type might serve that role.
- **Legal documents:** Neither attribute would reliably apply.

The heuristic is therefore dictated by which attributes are reliable and meaningful in the specific domain.

A notable example of a more general, domain-agnostic blocking heuristic is **Locality Sensitive Hashing (LSH)**. It groups similar pairs into candidate buckets based on hash collisions, this was covered in Chapter 3 of this course.

1.2.2 Comparison

For each remaining pair $(e_1, e_2) \in S$, the comparison step quantifies their similarity across d attributes using a set of similarity functions $sim_1, sim_2, \dots, sim_d$. The i -th coordinate of the resulting point p_{e_1, e_2} equals $sim_i(e_1, e_2)$, where a higher value indicates greater similarity on that attribute. The choice of similarity function depends on the attribute type:

- For numerical attributes such as **p-price**, one may use $-|e_1.\text{price} - e_2.\text{price}|$, where the negation ensures consistency with the convention that higher values reflect greater similarity.
- For text attributes such as **p-name** and **p-description**, functions such as edit distance, Jaccard similarity, or cosine similarity may be employed.

This process yields a d -dimensional set of points $P = \{p_{e_1, e_2} \mid (e_1, e_2) \in S\}$ which is handed over to the Matching step.

Example 1.1.

1.2.3 Matching

The entity matching objective is now made a binary classification task of assigning each point $p_{e_1, e_2} \in P$ to its correct ground truth label, which is $+1$ for a matching pair and -1 otherwise. While the ultimate approach to this would be with human judgement, this paper introduces **Monotone Classification** as a way to automate the process while ensuring a high degree of labelling accuracy.

1.3 Monotone Classification

To minimize the reliance on human judgement, we instead want to use a classifier to correctly determine the correct label to each of the given points from P in d dimensional space. So first, we can formally define our classifier:

1.3.1 The Importance of Monotonicity

1.3.2 Active Classification

2 RPE Algorithm

2.1 The Algorithm

2.2 Why It Works

2.3 Guarantees and Limitations

3 Coresets (RCC)

3.1 Why RPE Isn't Enough

3.2 The Coreset Technique

3.3 Achieving the $1+\varepsilon$ Guarantee

4 Proofs and Lower Bounds

4.1 Lower Bound Results

4.2 Near-Optimality of the Algorithms

4.3 Mathematical Machinery

Summary

References

- [1] Y. Tao, “Monotone Classification,” *PODS’18*.
- [2] Y. Tao, “Monotone Classification (Extended),” *PODS’21*.